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Nuclear Fission Plant
Sustainable Nuclear Energy

Assignment Two Fase Flow

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1 General introduction

In the nuclear engineering field, a deep understanding of flow patterns and two-phase flow behavior is fundamental, as it represents a crucial aspect in the thermo-hydraulic design of fission reactors. The simultaneous presence of gas and liquid phases gives rise to complex phenomena such as non-uniform velocity distribution, pressure drops, and heat transfer mechanisms within the piping system.

This laboratory activity aims to investigate experimentally the flow regime and the pressure drop in a vertical plexiglass pipe carrying a two-phase upward flow of water and air. The tests were carried out by varying the air flow rate through dedicated rotameters and adjusting the water flow rate by selecting different calibrated diaphragms mounted on parallel water lines.

The main objectives of the activity are:

- to simulate an upward two-phase flow under controlled laboratory conditions;
- to directly observe and classify the flow patterns (bubbly, slug, churn, annular);
- to evaluate the average void fraction in the test section through the quick-closing valve method;
- to measure the pressure difference along the pipe using differential pressure transducers;
- to compare the experimental results with theoretical models and correlations.

The experimental data are analyzed and compared against established theoretical tools: flow regime maps (Hewitt-Roberts and Taitel-Duckler), correlations for void fraction estimation (Homogeneous, Zivi, Chisholm, CISE, Drift Flux) and models for pressure drop evaluation (elevation and friction components using Homogeneous and Friedel approaches). This comparison provides insight into the accuracy and applicability of simplified models for engineering predictions in multiphase nuclear systems.

2 Experimental Setup

2.1 Facility description

The experimental facility consists, as we can see, in Figure 1 of a vertical plexiglass pipe with an internal diameter of 26 mm, designed to allow the upward flow of a water air mixture. At the inlet of the pipe, a porous bronze mixing device ensures homogeneous mixing of the two phases. Fast closing valves are positioned at the top and bottom of the test section, enabling the isolation and mass measurement of the trapped liquid for void fraction evaluation.

At the outlet, the two phase mixture flows into a vertical separator, which splits the water and air; the water is returned to the tank and the air is released at the top of the circuit. The entire system is part of a closed loop arrangement with a recirculating water tank and a compressed air line.



Figure 1: Experimental setup: vertical test section with rotameters, diaphragms, and pressure taps.

2.2 Instrumentation and Data Acquisition

The laboratory is equipped with:

- PT100 thermoresistance, used for the measurement of water temperature, located upstream the set of diaphragms.
- Three differential diaphragms for measuring the water flow rate.
- Three differential rotameters, used to measure the air flow rate, with ranges of 0.2–1.2, 0.45–4.2, and 1.5–17.0 Nm^3/h
- Relative and absolute pressure transducer to determine the varying pressure in different points in the system.

2.3 Working Fluids and Flow Conditions

The working fluids used were compressed air and water at ambient temperature, both recirculated within a closed loop system.

The air flow rate was varied by selecting the appropriate rotameter range and adjusting the upstream pressure, with volumetric readings subsequently corrected to account for actual pressure conditions. The water flow rate was controlled by manually selecting among three calibrated diaphragms small and medium. The large diaphragm was not used, as the medium one provided pressure drops within the most accurate operating range of the differential pressure transducer, avoiding low end values where measurement uncertainty increases. Each diaphragm corresponds to a different measurement range, and the water flow rate was evaluated using specific calibration correlations based on the measured pressure drop.

By combining different air and water flow rates, a wide spectrum of two phase flow regimes was reproduced within the vertical pipe, ranging from bubbly to annular flow.

3 Flow pattern Classification

In order to represent the experimental points on the theoretical maps (Hewitt-Roberts and Taitel-Duckler), it was necessary to compute the superficial velocities of both gas and liquid in the vertical pipe.

The flow pattern, or flow regime, describes the spatial distribution and interaction between the liquid and gas phases inside the pipe. In vertical two-phase upward flow, several distinct flow regimes can be identified depending on the operating conditions, such as fluid velocities, densities, and surface tension.

- **Bubbly flow:** characterized by small gas bubbles dispersed in the continuous liquid phase, typically at low gas flow rates.
- **Slug flow:** large bullet-shaped gas bubbles alternate with liquid slugs; appears as gas velocity increases.
- **Churn flow:** chaotic and oscillating flow where liquid and gas phases are intensely mixed, often in transition between slug and annular flow.
- **Annular flow:** the gas core flows through the center of the pipe, surrounded by a thin liquid film along the walls, typical at high gas velocities.

Each of these patterns has a different impact on heat and mass transfer, pressure drop, and void fraction, making their identification crucial for accurate modeling in thermal-hydraulic analyses. The theoretical maps used in this work define boundaries between these regimes as functions of the superficial velocities or their derived dimensionless combinations.

Air Flow Rate through Rotameters:

The air flow rate is measured using rotameters calibrated in standard condition, 20 °C at 1 atm. Since the values provided by the rotameters are expressed in Nm³/h under standard conditions (20°C, 1 atm), it was necessary to correct them according to the actual inlet pressure measured by the manometer. The mass flow rate of air under the actual operating conditions was then computed using the following expression:

$$W_e = 1.204 \cdot \sqrt{\frac{p_{e,rel} + 1.013}{1.013}} \cdot \frac{Q_s}{3600} \quad (1)$$

in which W_e is the actual air mass flow rate [kg/s], $p_{e,rel}$ is the relative pressure upstream of the rotameter [bar], Q_s is the volumetric flow rate under standard conditions [Nm³/h] and 1.204 is the air density at 20°C and 1 atm [kg/m³].

By knowing the air mass flow rate, it has been computed the gas velocity (j_G) as follows:

$$V_{air} = \frac{W_e}{\rho_g} \longrightarrow j_G = \frac{V_{air}}{Area} \quad (2)$$

Water Flow Rate through diaphragm:

The water mass flow rate was determined based on the pressure drop measured across calibrated diaphragms. Due to the specific geometric configuration of these devices, the

standard ISO/CNR correlations could not be applied. Instead, empirical coefficients obtained through laboratory calibration were used. The flow rate was computed as:

$$W = \alpha_{MQ} \cdot \frac{\pi d^2}{4} \cdot \sqrt{2\rho\Delta p} + B \quad (3)$$

In which α_{MQ} and B represent two constants that depend by the chosen diaphragm (Table 1). The large diaphragm was not employed, as it produced pressure drops that were too small, resulting in less reliable measurements.

Diaphragm	D (m)	d (m)	$\beta=d/D$	α_{MQ}	B
small	0.0168	0.0084	0.5	0.650729	0.01415274
medium	0.0212	0.0153	0.7217	0.728193	0.00375718
large	0.034	0.0255	0.75	0.773353	0.00986774

Table 1: Diaphragm constants

In the Matlab implementation, each test point was processed within a loop. The diaphragm type was identified through a conditional check, and the corresponding coefficients α_{MQ} , B , and diameter d were assigned accordingly. The pressure drop Δp , initially measured in mbar, was converted into Pascal before being inserted into the equation. An error-handling instruction was included to detect invalid diaphragm labels.

Finally, the superficial velocity of the liquid was calculated using:

$$j_L = \frac{W}{\rho_L \cdot Area} \quad (4)$$

3.1 Hewitt- Roberts Flow Pattern Map

To compare the experimentally observed flow patterns with theoretical predictions, the Hewitt-Roberts map was used. This empirical map is specifically developed for vertical upward two phase flows and allows classification of flow regimes such as bubbly, slug, churn and annular, based on the superficial velocities of the gas and liquid phases.

The map plots the following two dimensionless parameters:

$$X = \rho_G \cdot j_G^2 \quad \left[\frac{\text{kg}}{\text{m}^2 \cdot \text{s}^2} \right], \quad Y = \rho_L \cdot j_L^2 \quad \left[\frac{\text{kg}}{\text{m}^2 \cdot \text{s}^2} \right] \quad (5)$$

where ρ_G , ρ_L are the densities of the gas and liquid phases.

Each experimental point was plotted using the computed X and Y values and annotated with a letter indicating the visually observed flow regime: B = Bubbly, S = Slug, C = Churn, A = Annular.

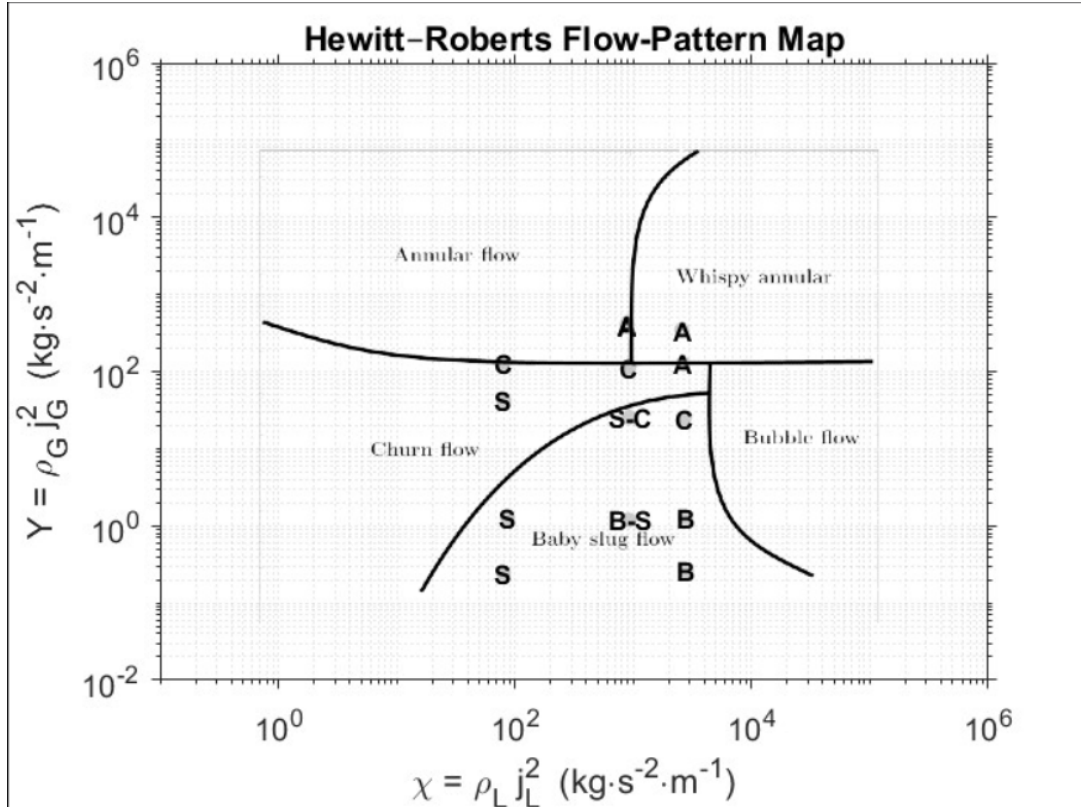


Figure 2: Hewitt-Roberts map with experimental data points and observed flow patterns.

In Figure 2 the map shows five main flow regimes: bubble, bubbly slug, slug, churn, and annular flow, with transitions occurring as a function of gas and liquid fluxes. However, the observed Bubbly flow points fall outside the theoretical bubble flow region, instead, they lie within the bubbly slug flow area, indicating a discrepancy between visual classification and theoretical boundaries. Similarly, some Slug–Churn (S–C) and Churn (C) points overlap near transition zones, confirming the difficulty in drawing sharp borders between regimes in practice. The discrepancy may originate from an uncertainty or subjectivity in visual flow regime identification, especially in borderline or rapidly changing flow conditions or a limitations of the map, which is based on a specific experimental setup that might not fully correspond to the test conditions.

3.2 Taitel-Duckler Flow Pattern Map

Transition Annular and Slug/Churn flow:

In this region of the Taitel-Duckler map, the transition from annular to slug/churn flow is identified through a theoretical boundary derived from momentum balance considerations. The coordinates X_t and Y_t , which are functions of the superficial velocities and fluid properties, allow the placement of experimental conditions on the map. The purpose is to verify whether the experimentally observed flow regime matches the one predicted by the model in this transition zone. The transition between annular and slug/churn flow is identified using the following dimensionless parameters:

$$X_t = \left(\frac{\left(\frac{dp}{dz} \right)_L}{\left(\frac{dp}{dz} \right)_G} \right)^{1/2} \quad (6)$$

$$Y_t = \frac{j_G \cdot \sqrt{\rho_G}}{[g \cdot (\rho_L - \rho_G) \cdot \sigma]^{1/4}} \quad (7)$$

Where j_G , j_L are superficial velocities of gas and liquid phases; ρ_G , ρ_L are the densities of gas and liquid; σ is the surface tension; (g) is the gravitational acceleration; $(\frac{dp}{dz})_L = \rho_L j_L^2$ and $(\frac{dp}{dz})_G = \rho_G j_G^2$ are the estimated pressure gradients for liquid and gas phases.

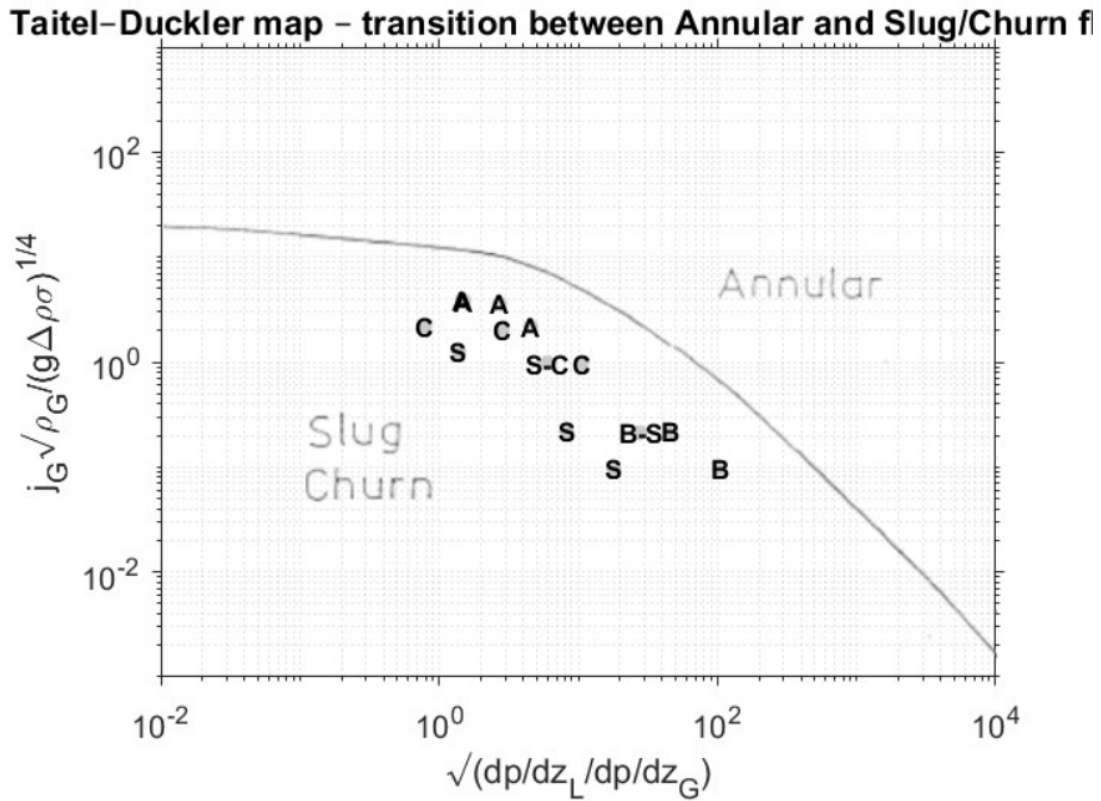


Figure 3: Transition annular to slug/churn with experimental data points and observed flow patterns.

In Figure 3 the map shows most experimental points fall reasonably close to the expected transition curve, even though some points classified as "Annular" (A) appear below the transition boundary, in the slug/churn region. This discrepancy could be caused from a non perfect visual interpretation. Instead the slug churn points are in the correct position, confirming the model's accuracy in predicting this flow regime transition.

Transition Slug and Churn flow:

The transition between slug and churn flow can be represented on a map where the horizontal axis is the dimensionless parameter $J/\sqrt{gD}$ and the vertical axis is the liquid velocity fraction β .

The dimensionless coordinates used to place the experimental points on this map are calculated as follows:

$$\beta = \frac{j_G}{j_L + j_G} \quad (8)$$

$$X = \frac{j}{\sqrt{gD}} \quad (9)$$

The terms j_L and j_G are the superficial velocities of liquid and gas, respectively; g is the gravitational acceleration and D is the internal diameter of the pipe.

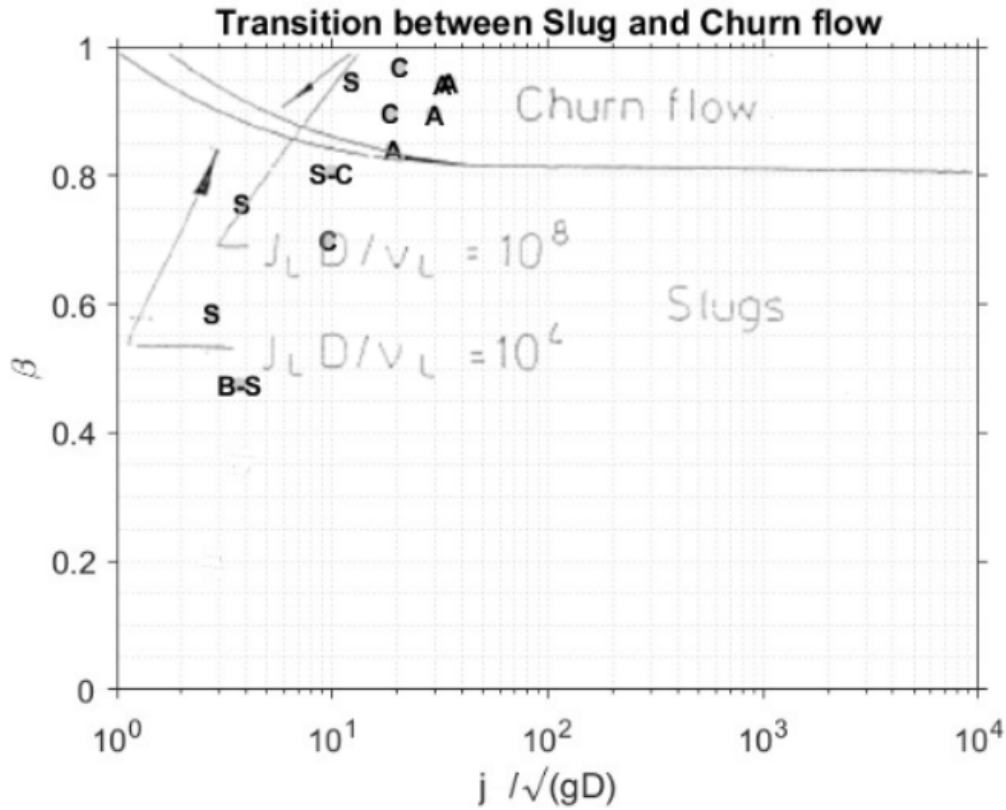


Figure 4: Transition slug to churn map with experimental data points and observed flow patterns.

In Figure 4 the map shows that the experimental points associated with slug and churn flow regimes fall correctly within the regions predicted by the model. Although one slug point appears within the churn region, this slight deviation can be attributed to experimental uncertainties or transitional flow behavior.

Transition Bubbly and Slug/Churn flow:

This map allows the identification of the transition between bubbly flow and slug/churn flow in vertical upward two-phase systems. The transition depends on the ratio between the liquid and gas superficial velocities and on the balance between gas inertia and the stabilizing effects of gravity and surface tension.

The dimensionless coordinates used to plot the experimental points are:

$$X = \frac{j_G \cdot \sqrt{\rho_L}}{[g \cdot (\rho_L - \rho_G) \cdot \sigma]^{1/4}} \quad (10)$$

$$Y = \frac{j_L}{j_G} \quad (11)$$

Where j is the superficial velocities and ρ is the density, in which the subscript L and G indicate the liquid or gas phase; σ represents the surface tension and g corresponds to the gravitational acceleration.

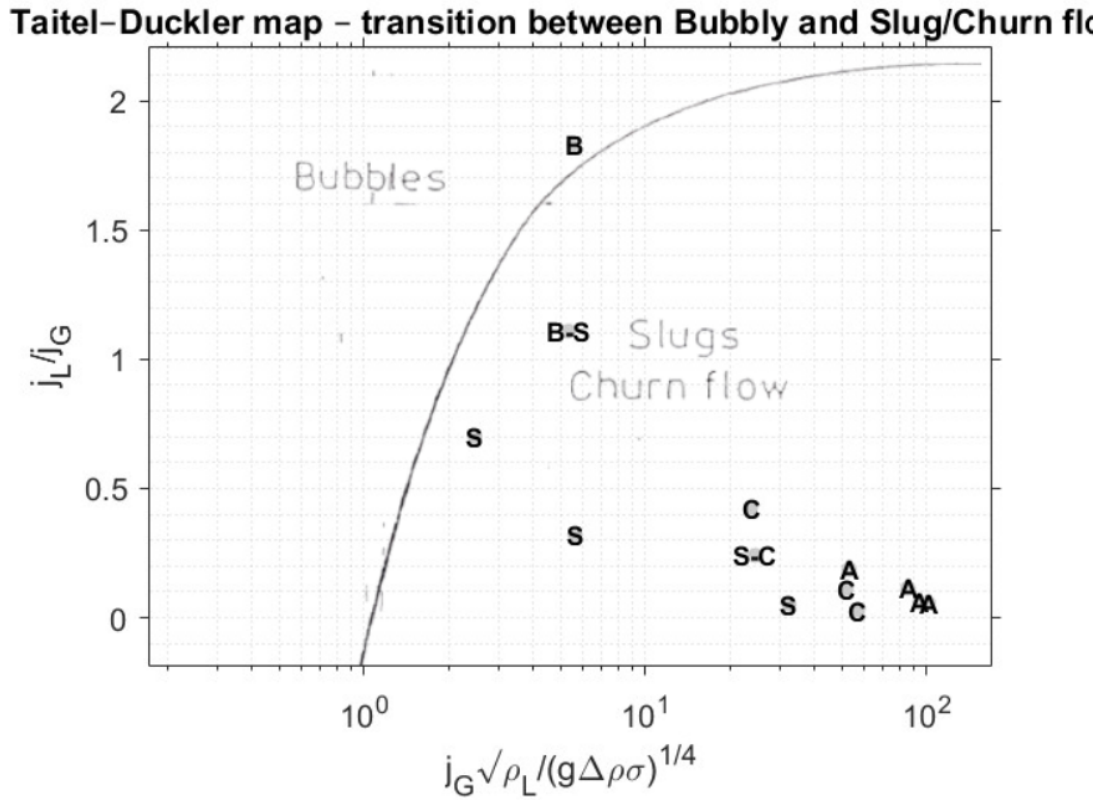


Figure 5: transition bubbly to slug/churn with experimental data points and observed flow patterns.

In Figure 5 the map allows to identify the transition to slug or churn flow as the gas flow rate increases relative to the liquid phase. Experimental points were calculated using the expressions (10) and (11) and compared with the theoretical boundary curve separating bubbly and slug/churn flow regimes. In general, the experimental points are well positioned because the majority of bubbly flows fall above the curve, while slug and churn flows fall below it. Just one point named as B–S appears slightly below the boundary, but being a transitional flow, its location near the interface is consistent with its nature and does not indicate a significant discrepancy.

4 Void Fraction Evaluation

The void fraction, denoted as α , is defined as the volumetric fraction of the gas phase within a two-phase mixture. It is a key parameter in the analysis of two-phase flow, as it directly influences fluid properties such as density, momentum, pressure drops, and heat transfer. From a volumetric standpoint:

$$\alpha = \frac{V_G}{V_G + V_L} \quad (12)$$

Where V_G and V_L represent the volumes occupied by the gas and liquid phases, respectively, in the control volume.

4.1 Experimental Evaluation of Void Fraction

The experimental void fraction was evaluated by isolating the two-phase mixture inside the test section using fast-closing valves and then measuring the amount of liquid contained within. The total mass of liquid in the test section when completely filled with water (i.e., in the absence of air) is:

$$M_{l0} = M_{\text{drain, full}} + M_{\text{res}} = 0.898 + 0.116 = 1.014 \text{ kg} \quad (13)$$

In the two-phase tests, the liquid mass is:

$$M_l = M_{\text{drain}} + M_{\text{res}} \quad (14)$$

The void fraction is then calculated as:

$$\alpha_{\text{exp}} = 1 - \frac{M_l}{M_{l0}} \quad (15)$$

4.2 Theoretical Models for Void Fraction

To compare with the experimental results, various theoretical correlations were implemented. Each model considers different assumptions regarding slip velocity and interfacial effects:

Homogeneous Model

Assumes no slip between phases:

$$\alpha_{\text{hom}} = \frac{1}{1 + \left(\frac{1-x}{x}\right) \cdot \frac{\rho_G}{\rho_L}} \quad (16)$$

Zivi Correlation

Introduces a slip ratio based on the density ratio:

$$S = \left(\frac{\rho_L}{\rho_G}\right)^{1/3}, \quad \alpha_{\text{Zivi}} = \frac{1}{1 + \left(\frac{1-x}{x}\right) \cdot \frac{\rho_G}{\rho_L} \cdot S} \quad (17)$$

Chisholm Correlation

Uses a slip ratio formulation derived from empirical observations:

$$S = \sqrt{1 - x \left(1 - \frac{\rho_L}{\rho_G}\right)}, \quad \alpha_{\text{Ch}} = \frac{1}{1 + \left(\frac{1-x}{x}\right) \cdot \frac{\rho_G}{\rho_L} \cdot S} \quad (18)$$

CISE Correlation

Provides improved accuracy for intermediate flow regimes:

$$\beta = \frac{\rho_L x}{\rho_L x + \rho_G (1 - x)} \quad (19)$$

$$y = \frac{1 - \beta}{1 + \beta} \quad (20)$$

$$Re = \frac{\rho_L j_L D}{\mu_L}, \quad We_{CISE} = \frac{\rho_L j_L^2 D}{\sigma} \quad (21)$$

$$E_1 = 1.578 \cdot Re^{-0.19} \left(\frac{\rho_L}{\rho_G} \right)^{0.22} \quad (22)$$

$$E_2 = 0.0273 \cdot We_{CISE}^{-0.51} \left(\frac{\rho_L}{\rho_G} \right)^{-0.08} \quad (23)$$

$$S = 1 + E_1 \left(\frac{y}{1 + y E_2} - y E_2 \right)^{0.5}, \quad \alpha_{CISE} = \frac{1}{1 + \left(\frac{1-x}{x} \right) \cdot \frac{\rho_G}{\rho_L} \cdot S} \quad (24)$$

Note: In this context, We_{CISE} indicates the Weber number used in the CISE correlation, and should not be confused with the symbol We_e used elsewhere in the code to denote the air mass flow rate.

Drift Flux Model

Accounts for slip velocity between phases and is particularly suited for vertical upward flows:

$$\alpha = \frac{j_G}{C_0 j + u_{gj}} \quad (25)$$

with:

- Bubbly flow: $C_0 = 1.13$, $u_{gj} = 1.41 \left(\frac{\sigma(\rho_L - \rho_G)}{\rho_L^2} \right)^{0.25}$
- Plug flow: $C_0 = 1.2$, $u_{gj} = 0.35 \cdot \sqrt{\frac{(\rho_L - \rho_G)gD}{\rho_L}}$

4.3 Comparison with Experimental Data

All the above correlations were implemented in Matlab and theoretical predictions were compared against the experimentally obtained void fraction α_{exp} . The results are presented in the following parity plot, where the ideal agreement line ($y = x$) is shown along with $\pm 20\%$ deviation bands. This visual comparison allows a straightforward assessment of the performance of each model under the tested flow conditions.

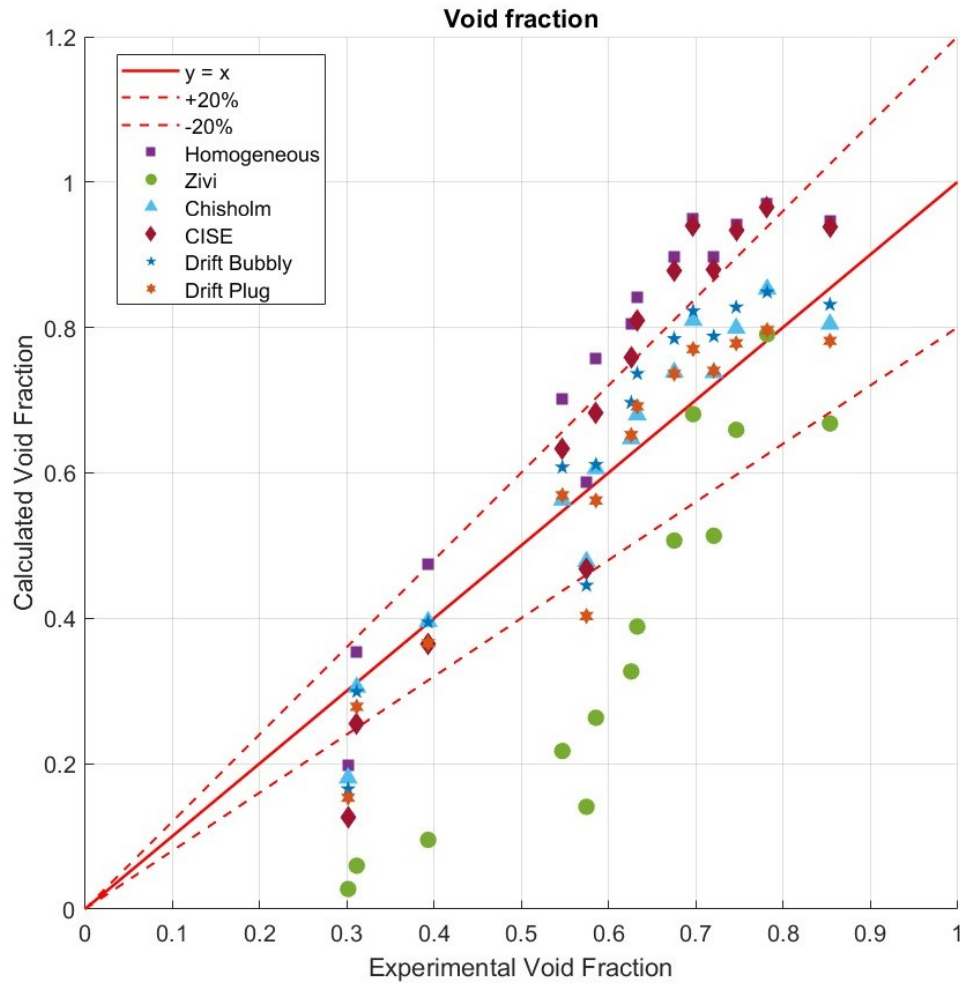


Figure 6: Comparison experimental and theoretical void fraction

As shown in Figure 6, the Drift bubbly and plug correlations show the best overall agreement with the experimental data, with the majority of its predicted points falling within the $\pm 20\%$ deviation band. The Chisholm correlation demonstrates moderate accuracy. Instead, the Homogeneous and Cise models tend to overestimate the void fraction, especially at higher values of α , while the Zivi correlation significantly underpredicts the void fraction across almost the entire dataset, confirming its limited applicability in the present experimental conditions.

5 Experimental Pressure Drop

The pressure drop across the vertical test section was experimentally measured using a Rosemount differential pressure transducer. As described in the documentation, this transducer can only measure positive pressure differences; therefore, the connection of the pressure taps to the "High" and "Low" chambers was adapted based on the expected pressure gradient during the test.

In particular, depending on the flow regime and the magnitude of the frictional and hydrostatic pressure terms, the measured differential pressure may either represent $p_C - p_D$ or $p_D - p_C$. Based on the physical configuration and using fluid statics (Bernoulli principle), the total pressure difference between points A and B in the test section is evaluated as:

- If $\Delta p_{\text{transd}} = p_C - p_D$:

$$\Delta p_{AB} = \bar{\rho}gh + \Delta p_{\text{frict}} = \Delta p_{\text{transd}} + \rho gh \quad (26)$$

- If $\Delta p_{\text{transd}} = p_D - p_C$:

$$\Delta p_{AB} = \bar{\rho}gh + \Delta p_{\text{frict}} = \rho gh - \Delta p_{\text{transd}} \quad (27)$$

where $\bar{\rho}$ is the average mixture density and $h = 1.5$ m is the vertical distance between the pressure taps.

5.1 Matlab Implementation and Data Handling

The experimental values were calculated in Matlab, taking into account the pressure difference type indicated in the dataset for each test. The variable `Pipe_pressure_drop` corresponds to the differential pressure value given by the transducer and depending on whether the connection was D-C or C-D, the total pressure drop Δp_{AB} was computed accordingly.

The following logic was applied:

- For type D-C: $\Delta p_{AB} = \rho gh - \Delta p_{\text{transd}}$
- For type C-D: $\Delta p_{AB} = \rho gh + \Delta p_{\text{transd}}$

Each pressure reading was converted from mbar to Pascal and recorded in a structured output table. For cases where the signal was unstable or fluctuated between positive and negative values preventing a reliable interpretation of flow direction a warning message was displayed in place of a numerical result.

5.2 Results and Visualization

The table below summarizes the evaluated experimental pressure drops for each test condition. Values range from approximately 4600 Pa to over 20000 Pa, depending on the

gas-liquid mixture and flow rates. Some tests showed unstable readings due to rapid flow regime transitions, which caused the transducer to oscillate between positive and negative differentials. These cases were flagged and excluded from the quantitative comparison with theoretical models.

6 Elevation Pressure Drop

The hydrostatic pressure drop in the vertical test section was evaluated using both the experimentally measured void fraction and the values estimated through theoretical models, namely the Homogeneous model, Chisholm correlation, CISE correlation, and the Drift Flux model (for both bubbly and plug flow assumptions).

The pressure drop due to elevation can be expressed as:

$$\Delta p_{\text{elev}} = \rho_{\text{avg}} \cdot g \cdot h \quad (28)$$

Where ρ_{avg} is the average density of the two-phase mixture, calculated as:

$$\rho_{\text{avg}} = \alpha \cdot \rho_G + (1 - \alpha) \cdot \rho_L \quad (29)$$

and $h = 1.5$ m is the vertical distance between the pressure taps.

6.1 Implementation and Results

The average mixture densities were computed in Matlab for each model using the corresponding void fraction values. These densities were then used to calculate the hydrostatic pressure drop Δp_{elev} . A comparison was made with the pressure drop obtained from the experimental void fraction.

The results are summarized in Table 2, which reports the hydrostatic pressure drop calculated via each correlation.

Experimental_E...	Elevation_Hom...	Elevation_Chis...	Elevation_CISE	Elevation_Drift...	Elevation_Drift...
6.27e+03	6.08e+03	7.69e+03	7.83e+03	1.13e+04	1.15e+04
6.11e+03	3.60e+03	5.81e+03	4.69e+03	8.86e+03	9.29e+03
4.47e+03	7.72e+02	2.82e+03	9.10e+02	3.99e+03	4.66e+03
3.22e+03	4.55e+02	2.18e+03	5.17e+02	3.11e+03	3.82e+03
2.17e+03	8.17e+02	2.89e+03	9.36e+02	2.54e+03	3.27e+03
8.93e+03	7.75e+03	8.90e+03	9.35e+03	9.03e+03	9.45e+03
5.51e+03	2.88e+03	5.21e+03	3.55e+03	4.55e+03	5.21e+03
4.79e+03	1.53e+03	3.87e+03	1.82e+03	3.24e+03	3.94e+03
3.75e+03	8.73e+02	2.98e+03	1.00e+03	2.59e+03	3.32e+03
1.03e+04	1.18e+04	1.21e+04	1.29e+04	1.12e+04	1.15e+04
1.01e+04	9.53e+03	1.02e+04	1.10e+04	8.89e+03	9.31e+03
6.67e+03	4.41e+03	6.45e+03	5.40e+03	4.63e+03	5.28e+03
5.41e+03	2.35e+03	4.73e+03	2.82e+03	3.20e+03	3.91e+03
4.13e+03	1.54e+03	3.87e+03	1.80e+03	2.68e+03	3.40e+03

Table 2: Elevation pressure changes between different correlation

From the results in Table 2, it appears that the elevation pressure drop calculated using the **Drift Flux model (bubbly)** is the most consistent with the experimental values. Its good performance is likely due to the empirical tuning of C_0 and u_{gj} for air–water flows in bubbly and slug regimes, which match the conditions of the present experiment. This leads to realistic void fraction estimates and thus accurate values of the mixture density.

The **CISE** model performs well in selected cases, but tends to overpredict the pressure drop when the void fraction is high. This is mainly due to its strong sensitivity to Reynolds and Weber numbers: at the lower Re and We of this setup, the slip correction becomes exaggerated, resulting in an overestimation of the average density.

The **Homogeneous model** shows larger deviations, especially in gas-dominated flows. Its no-slip assumption leads to incorrect void fractions and overestimated densities, causing a general underprediction of the elevation term.

The **Chisholm model** offers intermediate results. Despite its simplified formulation, it performs reasonably well in several tests—especially at moderate void fractions—and is less sensitive to flow conditions than CISE, making it a useful compromise between accuracy and simplicity.

These trends are consistent with physical expectations: increasing void fraction reduces the average density and thus the hydrostatic pressure drop. The comparison confirms that the accuracy of the elevation term depends strongly on the choice of void fraction model, especially under conditions with significant slip.

7 Frictional Pressure Drop

The frictional pressure drop Δp_{fric} was calculated using two different models:

- **Homogeneous model**, assuming no slip between phases:

$$\Delta p_{\text{fric,hom}} = \Delta p_{AB} - \rho_{\text{avg,hom}} \cdot g \cdot h \quad (30)$$

where the average density is calculated as:

$$\rho_{\text{avg,hom}} = \alpha_{\text{hom}} \cdot \rho_G + (1 - \alpha_{\text{hom}}) \cdot \rho_L \quad (31)$$

- **Friedel correlation**, which accounts for slip and complex phase interactions in vertical upward flows. The model is based on the liquid only pressure gradient, corrected with a multiplier:

$$\Delta p_{\text{fric,Friedel}} = \left(\frac{f_L \cdot \rho_L \cdot j_L^2}{2D} \right) \cdot \phi^2 \cdot h \quad (32)$$

where:

$$\phi^2 = E + \frac{3.24 \cdot F \cdot H}{Fr_k^{0.045} \cdot We_k^{0.035}} \quad (33)$$

with the following definitions:

$$E = (1 - x)^2 + x^2 \cdot \frac{\rho_L}{\rho_G} \cdot \frac{f_G}{f_L}, \quad F = x^{0.78} (1 - x)^{0.224} \quad (34)$$

$$H = \left(\frac{\rho_L}{\rho_G} \right)^{0.91} \left(\frac{\mu_G}{\mu_L} \right)^{0.19} \left(1 - \frac{\mu_G}{\mu_L} \right)^{0.7} \quad (35)$$

$$Fr_k = \frac{j^2}{gD\rho_k}, \quad We_k = \frac{j^2 D}{\sigma\rho_k}, \quad \rho_k = \left(\frac{x}{\rho_G} + \frac{1-x}{\rho_L} \right)^{-1} \quad (36)$$

Friction ΔP Homogeneous	Friction ΔP Friedel
47.7674	170.4926
86.7794	280.5427
391.6039	688.3970
676.4631	917.5355
3.0275e+03	4.5415e+03
318.2885	943.5667
866.7818	2.1733e+03
1.6377e+03	3.2161e+03
2.8374e+03	4.3261e+03
549.0759	1.1946e+03
673.2704	1.7523e+03
1.4315e+03	3.7485e+03
2.6534e+03	5.6826e+03
4.0004e+03	7.1919e+03

Table 3: Comparison of friction pressure drop between Homogeneous and Friedel

The comparison in Table 3 shows a clear discrepancy between the frictional pressure drop estimated using the Homogeneous model and that obtained with the Friedel correlation. The values computed with the Homogeneous model range from as low as 48 Pa up to approximately 4 kPa, increasing gradually with flow rate. This behavior is expected, as the model assumes a single phase formulation with no slip between phases, applying a classical friction factor to an average mixture density. While this approach is simple and computationally convenient, it significantly underestimates the actual pressure drop in two phase systems, especially at higher gas fractions.

On the other hand, the Friedel correlation yields consistently higher values, ranging from about 170 Pa to over 7 kPa. This is due to the introduction of a corrective multiplier ϕ^2 , which enhances the single phase liquid pressure gradient by accounting for the effects of phase interaction, flow regime transitions and velocity slip. The correction depends on Reynolds, Weber and Froude numbers, thus dynamically adapting to the test conditions.

Overall, the analysis demonstrates that the Homogeneous model, while useful for preliminary estimates, lacks the fidelity required for accurate prediction of frictional pressure drop in complex multiphase systems. The Friedel correlation proves to be a more reliable and comprehensive approach under the tested conditions.

8 Total Pressure Drop

In the following section, the pressure drop across the vertical test section is evaluated by summing the elevation and frictional contribution. The aim is to evaluate the impact of different modeling assumption on the prediction of total pressure drop in the two phase flow. The results of the total pressure drop estimation using different model combinations are shown in Figure 4. The four approaches considered are:

$$\Delta p_{\text{total}} = \Delta p_{\text{elev}} + \Delta p_{\text{fric}} \quad (37)$$

The four model combinations considered are:

1. Homogeneous model for both elevation and friction:

$$\rho_{\text{avg,hom}} = \alpha_{\text{hom}} \cdot \rho_G + (1 - \alpha_{\text{hom}}) \cdot \rho_L$$

$$\Delta p_{\text{elev,hom}} = \rho_{\text{avg,hom}} \cdot g \cdot h$$

$$\Delta p_{\text{fric,hom}} = \Delta p_{AB} - \Delta p_{\text{elev,hom}}$$

2. Friedel for friction, CISE for elevation:

The void fraction from the CISE correlation was used to compute:

$$\rho_{\text{avg,CISE}} = \alpha_{\text{CISE}} \cdot \rho_G + (1 - \alpha_{\text{CISE}}) \cdot \rho_L$$

$$\Delta p_{\text{elev,CISE}} = \rho_{\text{avg,CISE}} \cdot g \cdot h$$

The frictional component was computed using Friedel's correlation as:

$$\left(\frac{dp}{dz} \right)_{\text{Friedel}} = \left(\frac{f_L \cdot \rho_L \cdot j_L^2}{2D} \right) \cdot \phi^2 \quad \Rightarrow \quad \Delta p_{\text{fric,Friedel}} = \left(\frac{dp}{dz} \right)_{\text{Friedel}} \cdot h$$

3. Chisholm for friction, CISE for elevation:

The elevation component was the same as above, while the friction term was derived from:

$$\rho_{\text{avg,Chisholm}} = \alpha_{\text{Ch}} \cdot \rho_G + (1 - \alpha_{\text{Ch}}) \cdot \rho_L$$

$$\Delta p_{\text{fric,Chisholm}} = \Delta p_{AB} - \rho_{\text{avg,Chisholm}} \cdot g \cdot h$$

4. Friedel for friction, Drift Flux (bubbly) for elevation:

The void fraction from the Drift Flux model was used to compute:

$$\rho_{\text{avg,DF}} = \alpha_{\text{DF}} \cdot \rho_G + (1 - \alpha_{\text{DF}}) \cdot \rho_L$$

$$\Delta p_{\text{elev,DF}} = \rho_{\text{avg,DF}} \cdot g \cdot h$$

The friction term is again obtained from Friedel's correlation, as described above.

Total_Homogeneous	Total_Friedel_CISE	Total_Chisholm_CISE	Total_Friedel_DriftFlux
6.1294e+03	8.0026e+03	6.0373e+03	8.3419e+03
3.0857e+03	4.9707e+03	4.3606e+03	6.0127e+03
1.1639e+03	1.5988e+03	2.7262e+03	3.3152e+03
1.1312e+03	1.4343e+03	3.0143e+03	3.1631e+03
3.8443e+03	5.4780e+03	2.8975e+03	7.0379e+03
8.0660e+03	1.0290e+04	1.0871e+04	9.8577e+03
3.7461e+03	5.7271e+03	7.8752e+03	6.6516e+03
3.1662e+03	5.0355e+03	9.3571e+03	6.4033e+03
3.7101e+03	5.3279e+03	NaN	6.8770e+03
1.2368e+04	1.4059e+04	1.5049e+04	1.3480e+04
1.0199e+04	1.2718e+04	1.4742e+04	1.2072e+04
5.8444e+03	9.1530e+03	NaN	9.5335e+03
5.0037e+03	8.5007e+03	1.7062e+04	9.5767e+03
5.5384e+03	8.9919e+03	1.8707e+04	1.0332e+04

Table 4: Total pressure change with different couples of models

Table 4 shows the comparison of total pressure drops calculated using different combinations of friction and elevation models.

The **Homogeneous model** gives results that are generally within the expected range, but several values differ significantly from those of the other models and from what would be expected physically. This is mainly due to the assumption of no slip between phases, which leads to inaccurate estimates of the average mixture density, especially in tests with a high gas content.

The combination of **Friedel for friction and CISE for elevation** tends to produce lower total pressure drops, particularly at intermediate flow rates. This could be because the CISE correlation sometimes overestimates the void fraction, which lowers the average density and thus reduces the hydrostatic contribution. Nevertheless, this combination shows a regular and stable behavior across most cases and the values remain consistent.

Using **Chisholm for friction with CISE for elevation** generally leads to slightly higher total pressure drops compared to Friedel+CISE and at higher flow rates it aligns well with the results of Friedel+Drift Flux. This suggests that the Chisholm model balances better with the CISE elevation in some regimes. A missing value in the table is likely due to an undefined friction component in that case.

The **Friedel + Drift Flux** combination gives the highest total pressure drops in almost all tests, since the Drift Flux model usually predicts lower void fractions than CISE, which leads to a higher mixture density and a stronger hydrostatic contribution. Combined with the consistently high friction component from Friedel, this makes it the most conservative approach among those tested.

In conclusion, these results show how sensitive the total pressure drop estimation is to the choice of void fraction model, particularly for the elevation term. While Friedel consistently gives physically reliable friction predictions, it is the elevation model that drives the main differences. From a design perspective, **Friedel + Drift Flux** is likely the best option when a conservative estimate is preferred, while **Friedel + CISE** remains a valid and numerically stable alternative.

9 Pressure Drop comparison

The experimental total pressure drop, used as reference for comparison, was calculated as:

$$\Delta p_{\text{total,exp}} = \Delta p_{AB} - \rho_{\text{avg,exp}} \cdot g \cdot h$$

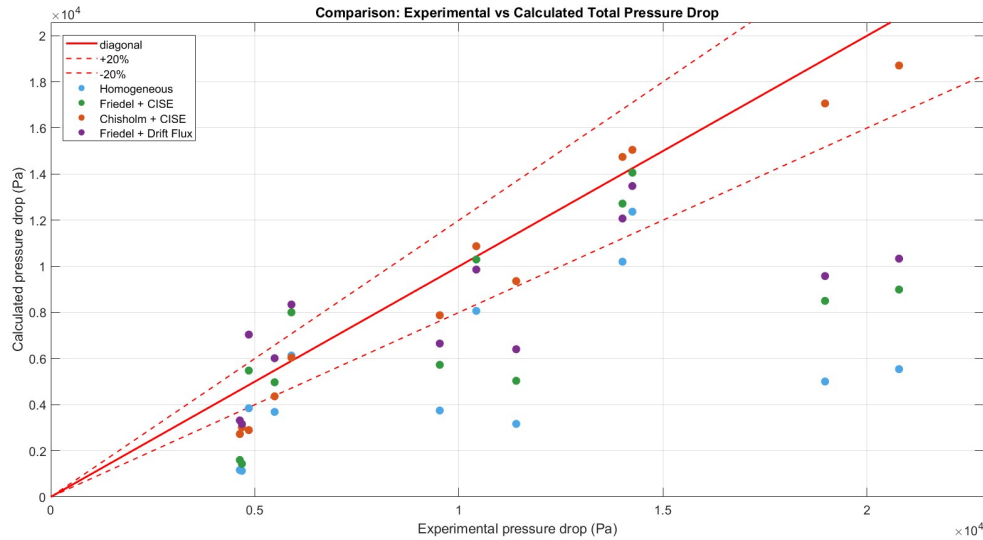


Figure 7: Comparison between experimental and calculated pressure drop

Figure 7 compares the calculated total pressure drops with the experimental values. The solid line indicates perfect match with the experimental values, while the dashed lines mark $\pm 20\%$ deviation.

From the plot, no model perfectly aligns with all experimental data, but some clear trends emerge:

- The **Homogeneous model** generally under-predicts the pressure drop. This is expected due to its simplified assumption of no slip, which leads to an under-estimation of the mixture density.
- The **Friedel + CISE** combination performs well overall. Most predictions fall within the $\pm 20\%$ band, showing consistent and balanced behavior across the test range.
- The **Chisholm + CISE** model shows quite good accuracy, with some results that fall under the band..
- The **Friedel + Drift Flux** model tends to underpredict the total pressure drop. This likely comes from the Drift Flux model estimating lower void fractions, which inflates the hydrostatic term and reduces the total.

In summary, the best agreement is observed for **Friedel + CISE**, followed by the **Chisholm + CISE** model. The choice of void fraction model especially for the elevation term strongly affects the final result, while the Friedel friction term remains a solid base across all combinations.

10 Conclusion

In conclusion, from this laboratory it has been possible to analyze the behavior of a two-phase flow in a vertical pipe and make a comparison with the experimental results with the theoretical correlation. In order to do so, it was not easy to identify the different patterns in the vertical pipe so made it challenging to achieve a good accuracy with respect to the theoretical models. The results discrepancy was also due to the fact that the two-phase mixture used in the laboratory was composed by water and air instead of steam.